

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Haresh Kumar N L	Reg. No.:	192425009
Section / Batch:	AI&ML	Date:	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Consider the concept map: Algorithm → Growth Function → Big-Omega → Lower Bound → Best-case Explain how a recursive algorithm can be visualized using a recursion tree and how costs are distributed across levels. Extend the map by including Tree Height and Branching Factor as influencing nodes. Justify how summing level costs leads to final complexity. Interpret how this map simplifies analysis of recursive algorithms.

Code of Conduct

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Haresh*

HOW REDUCING INPUT SIZE LEADS TO LOGARITHMIC GROWTH

- At every step, we reduce the input size by half.
- After k steps, $\text{size} = n/2^k$
- Stop when size becomes 1:
 $n/2^k = 1 \Rightarrow 2^k = n$
- Taking log on both sides: $k = \log_2 n$
- Number of steps $= O(\log n)$
- Time grows logarithmically.

SUMMARY

- Recursion tree shows the flow of recursive calls.
- Reducing input size by half gives $\log_2 n$ levels.
- Simple and effective way to analyze algorithms.

CONCEPT

- Recursion tree is a way to show how recursive calls are made.
- It helps us understand how the total cost is calculated.

RECURSION TREE VISUALIZATION & COST ANALYSIS

WHY BINARY SEARCH PERFORMS BETTER

- Eliminates half of the elements at every step.
- Requires far fewer comparisons.
- Time grows as $O(\log n)$ instead of $O(n)$.
- Faster and more efficient for large data.
- Saves time and computational resources.

FACTORS INFLUENCING THE RECURSION TREE

TREE HEIGHT (n)

- Height = number of levels from root to leaves.
- Each level reduces size by half.
 $n = \log_2 n$
- More height means more work.

BRANCHING FACTOR (b)

- Number of recursive calls at each node.
- If each node splits into b parts, each level has b^i nodes.
- Larger b means more work per level.

HOW RECURSION TREE SIMPLIFIES ANALYSIS

- Breaks the problem into levels.
- Helps us see how work is distributed.
- Makes it easy to calculate total cost.
- Useful for divide and conquer problems like binary search, merge sort.

APPLICATIONS

- Searching in databases
- Dictionary lookups
- Phone contacts
- System libraries
- Event scheduling

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____